

LI Databases and Web programming

Lab sheet Week 2 Solution

Scenario: To-Do Application

You are asked to design the database for a **To-Do Item Application**.

The application allows users to manage their **tasks** (to-do items), organize them by **projects**, and assign ownership. Each task also has a **priority**, a **status**, and can be linked to a **context**.

Here are the main details:

- A **To-Do Item** has:
 - a unique identifier,
 - a text description,
 - a due date, and
 - a completion date.
- Each **To-Do Item** belongs to one **Owner**.
 - An **Owner** has:
 - an ID,
 - a first name, and
 - a surname.
- Each **To-Do Item** is linked to a **Project**.
 - A **Project** has:
 - a project ID,
 - a project name, and
 - a budget.
- Each **To-Do Item** is associated with a **Context**.
 - A **Context** has:
 - a context ID, and
 - a description.
- Each **To-Do Item** also has a **Priority** (e.g., High, Medium, Low).
- Each **To-Do Item** also has a **Status** (e.g., Pending, In Progress, Completed).

Task 1: Extract Entities

From the scenario, list the **entities** and their **attributes**.
Write them in this format:

- Entity: Attributes

Task 3: Write the JDL File

After identifying the entities and relationships, write a **JDL model** for this system.

Task 4: Visualize the ER Diagram

- Open **JDL Studio** (<https://start.jhipster.tech/jdl-studio/>).
- Paste your JDL code.
- Generate and **draw the ER diagram**.

Solution

Task 1

Entity: ToDoItem

Attributes: id, description, dueDate, completionDate, priority, status

Entity: Owner

Attributes: id, firstName, surname

Entity: Project

Attributes: id, name, budget

Entity: Context

Attributes: id, description

Task 2-3:

```
entity Owner {  
    firstName String required,  
    surname String required  
}
```

```
entity Project {  
    name String required,  
    budget BigDecimal  
}
```

```
entity Context {  
    description String required  
}
```

```
entity ToDoItem {  
    description String required,  
    dueDate Instant,  
    completionDate Instant,  
    priority ToDoPriority required,  
    status ToDoStatus required  
}
```

```
enum ToDoPriority {  
    High, Medium, Low  
}
```

```
enum ToDoStatus {  
    Pending, InProgress, Completed
```

```
}
```

```
relationship ManyToOne {  
    ToDoItem{owner} to Owner,  
    ToDoItem{project} to Project,  
    ToDoItem{context} to Context  
}
```

Task 4: Visualize in JDL Studio

Copy the above JDL code.

Go to <https://start.jhipster.tech/jdl-studio/>

Paste the code and generate the ER diagram.

